

MATH70102 Unsupervised Learning

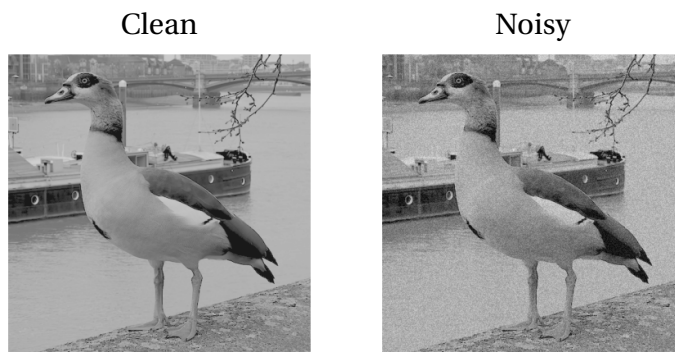
Coursework 1 (50%)

General rules — please read carefully:

- ★ This assignment must be attempted **individually** — your submission must be your own, unaided work. Candidates are prohibited from discussing assessed coursework, and must abide by Imperial College's rules regarding academic integrity and plagiarism. Violations will be treated as an examination offence. Enabling other candidates to plagiarise your work constitutes an examination offence. To ensure quality assurance is maintained, departments may choose to invite a random selection of students to an “authenticity interview” on their submitted assessments.
 - ★ Try to solve **all 4 problems**.
 - ★ You are expected to spend 1–2 workdays on this coursework.
 - ★ You may use any software as long as it is publicly available (either commercially or for free).
 - ★ Present your results in a **single PDF document with no more than 12 pages**, code included. You may generate the PDF from a Jupyter notebook, but the following restrictions apply:
 - PDFs where the bulk of text is rendered as an image will not be accepted;
 - The PDF must have a standard page size (A4 or letter) — in particular, PDFs where the entire notebook is rendered as a single (long) page will not be accepted.
 - ★ Any references used besides the course material must be cited.
 - ★ Use of **generative AI tools is permitted** but the following caveats apply. If you have used generative AI tools for the assignment, you must include an **acknowledgement that specifies how they have been used**. Moreover, **marks may be deducted for excessive reliance on generative AI**.
 - ★ The coursework is due on **Wednesday, 3 June 2026, 9:00am BST**.
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Problem 1 [15 marks]

The pixel values of both clean and noisy 512×512 greyscale images depicting an Egyptian goose are provided in the files `egyptian_goose_clean.csv` and `egyptian_goose_noisy.csv`, respectively.



Select integers q that adequately cover the range $1, \dots, 512$, and for each such q , do the following:

- ★ Apply *principal component analysis* (PCA) to the *noisy* image by treating its rows or columns as samples in $\mathbb{R}^{512}$. Then reconstruct the image from the first q principal components.
- ★ Apply *non-negative matrix factorisation* (NMF) to the *noisy* image, factorising it in terms of $W \in \mathbb{R}_{\geq 0}^{512 \times q}$ and $H \in \mathbb{R}_{\geq 0}^{q \times 512}$. Then reconstruct the image via WH .

Finally, is it possible to find integers q for both PCA and NMF such that the reconstructed images are closer to the *clean* image in the Euclidean distance than the original noisy image? That is, are PCA and NMF able to perform denoising of the image?

Problem 2 [10 marks]

Great circle distances (in kilometres) between some major airports in the northern hemisphere are provided in the following table:

EWB	6223				
KEF	2151	4185			
LHR	981	5577	1900		
NRT	8734	10833	8844	9614	
YVR	7685	3908	5702	7601	7521
	CPH	EWB	KEF	LHR	NRT

Key: CPH = Copenhagen, EWB = Newark, KEF = Keflavík, LHR = London Heathrow, NRT = Tokyo Narita, YVR = Vancouver.

- Use this information to draw a map (mathematically, an embedding in $\mathbb{R}^2$) which displays these airports (and nothing else), representing the distances as faithfully as possible (in relative terms). The orientation of the map can be arbitrary. [4 marks]
- How accurate is your map in terms of distances? Why can it not be perfectly accurate? [2 marks]
- Suppose that you increase the dimensionality of the “map” — that is, embed the distances in $\mathbb{R}^d$ for $d > 2$ instead of $\mathbb{R}^2$. Does that result in higher accuracy? [4 marks]

Problem 3 [10 marks]

Construct three data sets $x_{1,1}, \dots, x_{1,n} \in \mathbb{R}$, $x_{2,1}, \dots, x_{2,n} \in \mathbb{R}$ and $x_{3,1}, \dots, x_{3,n} \in \mathbb{R}$, using simulated or real data, and their mixtures

$$y_{1,i} := a_{1,1}x_{1,i} + a_{1,2}x_{2,i} + a_{1,3}x_{3,i}, \quad i = 1, \dots, n,$$

$$y_{2,i} := a_{2,1}x_{1,i} + a_{2,2}x_{2,i} + a_{2,3}x_{3,i}, \quad i = 1, \dots, n,$$

$$y_{3,i} := a_{3,1}x_{1,i} + a_{3,2}x_{2,i} + a_{3,3}x_{3,i}, \quad i = 1, \dots, n,$$

with some weights $a_{i,j} \in \mathbb{R}$, $i, j = 1, \dots, 3$, such that you can demonstrate the following:

- ★ *Principal component analysis* (PCA) cannot recover the original data sets $x_{1,1}, \dots, x_{1,n}$, $x_{2,1}, \dots, x_{2,n}$ and $x_{3,1}, \dots, x_{3,n}$ from the mixtures $y_{1,1}, \dots, y_{1,n}$, $y_{2,1}, \dots, y_{2,n}$ and $y_{3,1}, \dots, y_{3,n}$.
- ★ By contrast, *independent component analysis* (ICA) recovers $x_{1,1}, \dots, x_{1,n}$, $x_{2,1}, \dots, x_{2,n}$ and $x_{3,1}, \dots, x_{3,n}$ from $y_{1,1}, \dots, y_{1,n}$, $y_{2,1}, \dots, y_{2,n}$ and $y_{3,1}, \dots, y_{3,n}$ up to reordering, scaling and some minor error.

Include a rationale for your construction.

Problem 4 [15 marks]

- (a) Simulate mutually independent random vectors $\mathbf{X}_1, \dots, \mathbf{X}_{10} \sim N_d(\mathbf{0}, I_d)$ for a suitable range of values of $d \geq 2$ to study the behaviour of the ratio

$$R := \frac{\max_{i \neq j} \|\mathbf{X}_i - \mathbf{X}_j\|}{\min_{i \neq j} \|\mathbf{X}_i - \mathbf{X}_j\|}$$

as $d \rightarrow \infty$. (Here $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^d$.) [5 marks]

- (b) Can you find a theoretical explanation for the behaviour of R as $d \rightarrow \infty$? [8 marks]
- (c) Comment briefly on what the behaviour of R as $d \rightarrow \infty$ has to do with the curse of dimensionality? [2 marks]